

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 3 – Comparative Analysis

|                  |                                                                                                                                   |               |                                          |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------|
| Course Code:     | DSA03                                                                                                                             | Course Title: | Natural Language Processing              |
| CO Assessed:     | CO4 – Precision                                                                                                                   | Assessment:   | Assessment Tool 3 – Comparative Analysis |
| Weightage:       | 30% of CIA                                                                                                                        |               |                                          |
| CO5 Statement:   | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) |               |                                          |
| Student Name:    | M.Rishitha                                                                                                                        | Reg. No.:     | 192424335                                |
| Section / Batch: | 2024 AIDS                                                                                                                         | Date:         | 01-09-2026                               |

### Code of Conduct

I M.Rishitha[192424335] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Rishitha

### Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

| Section / Topic                                         | Focus                                                                                                                                                                          | Question No. |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representing Meaning & Meaning Structure of Language    | Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words. | Q1           |
| Syntax-Driven Semantic Analysis                         | Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.              | Q2           |
| Lexemes and Their Senses & Word Sense Disambiguation    | Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.          | Q3           |
| Representing Linguistically Relevant Concepts           | Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.                            | Q4           |
| Information Retrieval & Syntax-Driven Semantic Analysis | Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.                          | Q5           |

## Annexure A – Comparative Analysis

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1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

## Rubrics for Calculation

| Criteria                | Weightage (%) | Level 4 (Exemplary)                                                           | Level 3 (Proficient)                            | Level 2 (Developing)                       | Level 1 (Beginning)                      |
|-------------------------|---------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| Scope of Items          | 20%           | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%           | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%           | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%           | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation             |
| Clarity of Presentation | 15%           | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |

| Q. No. / Criteria Level | Scope of Items | Comparison Depth | Use of Evidence | Critical Thinking | Clarity of Presentation | Sub. Total | Total % |
|-------------------------|----------------|------------------|-----------------|-------------------|-------------------------|------------|---------|
| 1                       | 4              | 4                | 4               | 4                 | 4                       | 20         | 92      |
| 2                       | 4              | 4                | 4               | 3                 | 3                       | 18         |         |
| 3                       | 4              | 4                | 4               | 3                 | 3                       | 18         |         |
| 4                       | 4              | 4                | 4               | 3                 | 3                       | 18         |         |
| 5                       | 4              | 4                | 4               | 3                 | 3                       | 18         |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |